

PROGRAM

SENDER

```
#include <stdio.h>
#include <sys/stat.h>
#include <sys/types.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    int fd;
    char * myfifo = "/tmp/myfifo";

    // Create the FIFO (Named Pipe)
    mkfifo(myfifo, 0666);

    printf("Sender: Writing to pipe...\n");
    // Open in Write mode
    fd = open(myfifo, O_WRONLY);

    // Write message
    char msg[] = "Hello from Sender!";
    write(fd, msg, strlen(msg)+1);

    printf("Sender: Message sent: %s\n", msg);

    close(fd);
    return 0;
}
```

RECEIVER

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/msg.h>

struct msg_buffer {
    long msg_type;
    char msg_text[100];
} message;

int main() {
    key_t key;
    int msgid;

    // Step 2: Generate Key
    key = ftok("progfile", 65);

    // Step 3: Open Queue
    msgid = msgget(key, 0666 | IPC_CREAT);

    // Step 4: Receive message
    msgrcv(msgid, &message, sizeof(message), 1, 0);

    // Step 5: Print
```

```
printf("Data Received is : %s\n", message.msg_text);

// Step 6: Remove Queue
msgctl(msgid, IPC_RMID, NULL);

return 0;
}
// Note: Create a dummy file named 'progfile' in the same folder before running.
```

OUTPUT

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ gcc 4b_sender.c -o 4b_sender
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ gcc 4b_receiver.c -o 4b_receiver
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ ./4b_sender & ./4b_receiver
[1] 9114
Receiver: Waiting for data...
Sender: Writing to pipe...
Receiver: Message received: Hello from Sender!
Sender: Message sent: Hello from Sender!
[1]+  Done                  ./4b_sender
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$
```